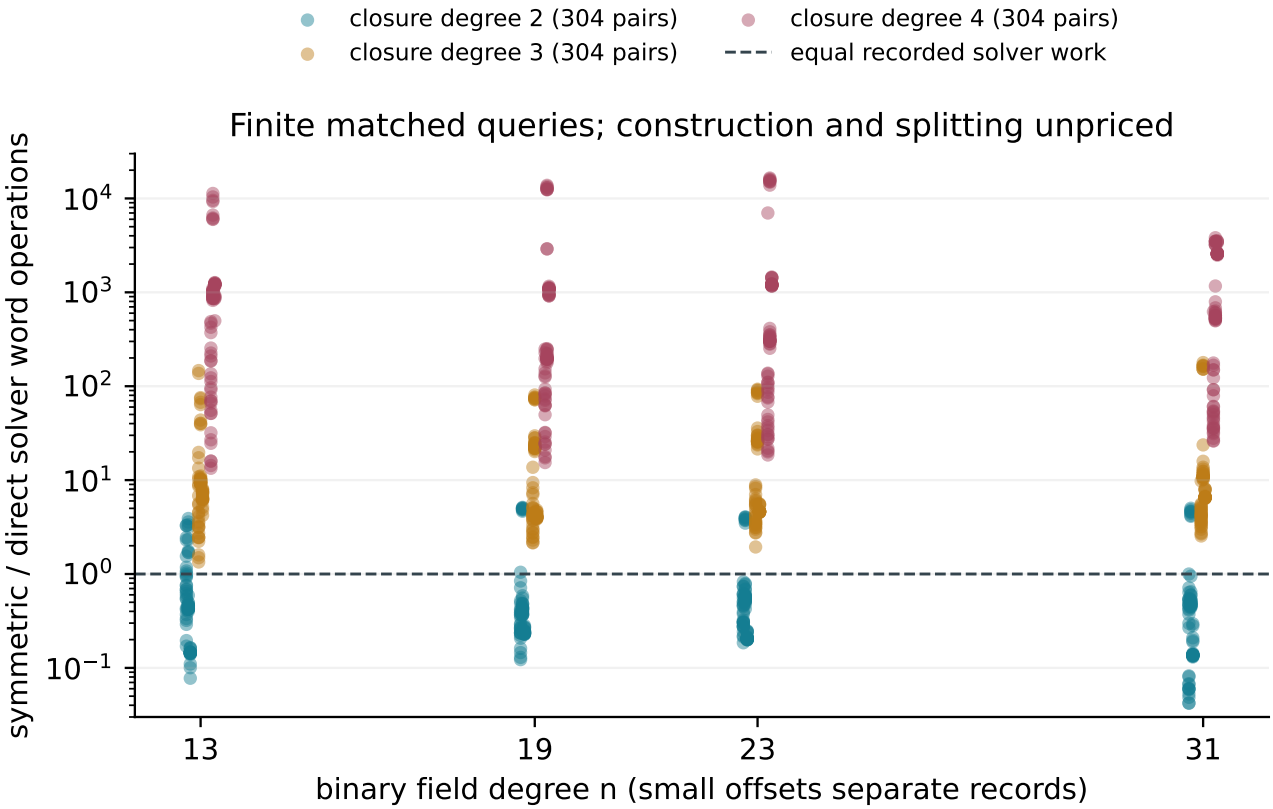
Native algebraic IC follow-up: measured stage results

Report date: 2026-10-25. Status: finite stage evidence; the wider campaign is partial.

Degree-two symmetric preprocessing has fewer recorded solver words in 263 of 304 completed paired queries. Degrees three and four have more in every completed pair. These are native count observations, not full-query or DLP speedups. The direct hybrid's input-degree floor may reach six.

Closure	raw rows	complete extractions	symmetric lower	ratio range
2	1152	768	263/304	0.04201–5.123
3	1152	891	0/304	1.342–178.5
4	1152	923	0/304	13.37–1.65e+04



All three panels use the same public targets and seeded bases at $n = 13, 19, 23, 31$, $\ell = 2, 3, 4$. Prefix duplicates are retained in raw data and excluded from the primary comparison. There are 336 distinct inputs, 304 completed matched pairs; 32 incomplete pairs are excluded and remain unknown. Direct group enumeration found no answer mismatch. The 8,100 symmetric algebraic roots in the completed queries yielded only four verified x-multisets; 8,096 roots failed split/lift.

Guessing, controls, and what remains

For the balanced three-set $[7, 15, 23]$ at the original ell=8 draw 0, all eight branches now saturate degree four without refuting. The rank is 129,221 on branches 0–3 and 129,473 on 4–7. This independent replay agrees with the upstream branch-0 result and fills seven previously unmeasured branches. It establishes failure of this set on this draw, not a global minimum.

The exact fixture oracle independently reproduces all seven original counts; all three SAT controls have checked Boolean witnesses. Small controls cover 2,304 direct assignments, every corresponding small symmetric root, repeated roots and 2,880 signed-lift cases. The u128 builder matches the library's actual equations at m=3–6; sparse and parallel fixed points match the dense controls.

The first-two-summand exchange is checked on all 308 coordinate equations of the seven original fixtures. Certificate `IDC1ha27a77e72057f9cb` also checks S3's ordinary integer-polynomial symmetry. Its formal-proof pointer is unset. The permutation equates degree verdicts for $[7, 23, 24]$ and $[15, 23, 24]$, not their costs.

Original-setup failures and internal-cap refusals are retained. The first structural outer shell ended with a syntax error after all 96 native commands had succeeded; its amended-on-disk runner caused the tail error. Subsequent panels execute frozen runner copies. No failed attempt is assigned zero cost.

No full-price IC gain or rho crossover is established. Complete DLP smoke, remaining guess draws, ell=9, actual N=58/60 continuations, sparse N=100, quartic replay/scaling, wide arithmetic and external-CAS/hardware controls remain separate obligations. The current quartic implementation explicitly refuses m=16–18. Source-pinned jobs and blocked backlog cards preserve them.

The owner's total additional cloud-compute ceiling is USD 100. No paid job had been launched when this report was prepared. A read-only CI access probe is separate from a compute allocation.

Evidence and scope

Native recomputation: `examples/ic_campaign_report_data.rs`. Raw values and audit pointers: `../../report-data.json`. Per-panel reports are siblings under `../`. Frozen requirements and amendments: `../../PROTOCOL.md`. Upstream references: fall-degree PR 2214; guessing PR 2429; cryptanalysis structured-base study. Solver construction, cubic splitting, lifting, physical conversion and the complete IC/rho pipeline are not collapsed into the plotted word ratios.